

16.1

2026-06-01

16.1 In Examples 16.2 and 16.4, we presented the linear probability and probit model estimates using an example of transportation choice. The logit model for the same example is $P(AUTO = 1) = \Lambda(\gamma_1 + \gamma_2 DTIME)$, where $\Lambda(\cdot)$ is the logistic cdf in equation (16.7). The logit model parameter estimates and their standard errors are

$$\begin{array}{ccc} \hat{\gamma}_1 + \hat{\gamma}_2 DTIME & = & -0.2376 + 0.5311 DTIME \\ (se) & & (0.7505) \quad (0.2064) \end{array} \quad (1)$$

EXAMPLE 16.4 | The Transportation Data: Probit

In Example 16.2, we estimated a linear probability model using the transportation data, *transport*. In this example, we carry out probit estimation. The probit model is $P(AUTO = 1) = \Phi(\beta_1 + \beta_2 DTIME)$. The maximum likelihood estimates of the parameters are

$$\begin{array}{ccc} \hat{\beta}_1 + \hat{\beta}_2 DTIME & = & -0.0644 + 0.3000 DTIME \\ (se) & & (0.3992) \quad (0.1029) \end{array}$$

The values in parentheses below the parameter estimates are estimated standard errors that are valid in large samples. These standard errors can be used to carry out hypothesis tests and construct interval estimates in the usual way, with the qualification that they are valid in large samples. The negative sign of $\hat{\beta}_1$ implies that when commuting times via bus and auto are equal so $DTIME = 0$, individuals have a bias against driving to work, relative to public transportation. The estimated probability of a person choosing to drive to work when $DTIME = 0$ is $\hat{P}(AUTO = 1 | DTIME = 0) = \Phi(-0.0644) = 0.4743$. The

positive sign of $\hat{\beta}_2$ indicates that an increase in public transportation travel time, relative to auto travel time, increases the probability that an individual will choose to drive to work, and this coefficient is statistically significant.

Suppose that we wish to estimate the marginal effect of increasing public transportation time, given that travel via public transportation currently takes 20 minutes longer than auto travel. Using (16.11),

$$\begin{aligned} \frac{dp}{dDTIME} &= \phi(\hat{\beta}_1 + \hat{\beta}_2 DTIME) \hat{\beta}_2 \\ &= \phi(-0.0644 + 0.3000 \times 2)(0.3000) \\ &= \phi(0.5355)(0.3000) = 0.3456 \times 0.3000 = 0.1037 \end{aligned}$$

For the probit probability model, an incremental (10-minute) increase in the travel time via public transportation increases the probability of travel via auto by approximately 0.1037, given that taking the bus already requires 20 minutes more travel time than driving.

The estimated parameters of the probit model can also be used to “predict” the behavior of an individual who must choose between auto and public transportation to travel to work. If an individual is faced with the situation that it takes 30 minutes longer to take public transportation than to drive to work, then the estimated probability that auto transportation will be selected is calculated using (16.12):

$$\begin{aligned} \hat{p} &= \Phi(\hat{\beta}_1 + \hat{\beta}_2 DTIME) = \Phi(-0.0644 + 0.3000 \times 3) \\ &= 0.7983 \end{aligned}$$

Since the estimated probability that the individual will choose to drive to work is 0.7983, which is greater than 0.5, we “predict” that when public transportation takes 30 minutes longer than driving to work, the individual will choose to drive.

a. Calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$.

Using the logit model, when $DTIME = 1$, the estimated index is

$$-0.2376 + 0.5311(1) = 0.2935. \quad (2)$$

Therefore, the estimated probability is

$$\hat{p} = \Lambda(0.2935) = \frac{e^{0.2935}}{1 + e^{0.2935}} \approx 0.573. \quad (3)$$

Thus, the estimated probability that a person will choose automobile transportation is approximately 0.573, or 57.3%.

-
- b. Using the probit model results in Example 16.4, calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$. How does this result compare to the logit estimate? [*Hint*: Recall that Statistical Table 1 gives cumulative probabilities for the standard normal distribution.]

Using the probit model from Example 16.4, when $DTIME = 1$, the estimated index is

$$-0.0644 + 0.3000(1) = 0.2356. \quad (4)$$

Therefore, the estimated probability is

$$\hat{p} = \Phi(0.2356) \approx 0.593. \quad (5)$$

Thus, the estimated probability that a person will choose automobile transportation is approximately 0.593, or 59.3%. This is slightly higher than the logit estimate of approximately 0.573.

-
- c. Using the logit model results, compute the estimated marginal effect of an increase in travel time of 10 minutes for an individual whose travel time is currently 30 minutes longer by bus (public transportation). Using the linear probability model results, compute the same marginal effect estimate. How do they compare?

Given that public transportation currently takes 30 minutes longer than driving, we have $DTIME = 3$. Using the logit model, the index is

$$-0.2376 + 0.5311(3) = 1.3557. \quad (6)$$

The logit marginal effect is

$$\frac{\partial p}{\partial DTIME} = \lambda(1.3557) * (0.5311), \quad (7)$$

where $\lambda(z) = \Lambda(z)[1 - \Lambda(z)]$. Since $\Lambda(1.3557) \approx 0.7951$,

$$\lambda(1.3557) = 0.7951(1 - 0.7951) \approx 0.1629. \quad (8)$$

Therefore,

$$\frac{\partial p}{\partial DTIME} \approx 0.1629 * (0.5311) = 0.0865. \quad (9)$$

Thus, a 10-minute increase in public transportation travel time increases the probability of choosing automobile transportation by approximately 0.0865, or 8.65 percentage points.

In the linear probability model, the marginal effect is simply the slope coefficient on $DTIME$, which is 0.0703. Therefore, the LPM estimate is 0.0703, or 7.03 percentage points. The LPM estimate is slightly smaller than the logit estimate, and unlike the logit model, it is constant for all values of $DTIME$.

-
- d. Using the logit model results, compute the estimated marginal effect of a decrease in travel time of 10 minutes for an individual whose travel time is currently 50 minutes longer by driving. Using the probit results, compute the same marginal effect estimate. How do they compare?

Since driving currently takes 50 minutes longer than public transportation, we have

$$D\text{TIME} = \frac{B\text{USTIME} - A\text{UTOTIME}}{10} = -5. \quad (10)$$

For the logit model, the index is

$$-0.2376 + 0.5311(-5) = -2.8931. \quad (11)$$

The logit marginal effect is

$$\Lambda(z)[1 - \Lambda(z)]\tilde{\gamma}_2. \quad (12)$$

Since $\Lambda(-2.8931) \approx 0.0525$,

$$\Lambda(z)[1 - \Lambda(z)]\tilde{\gamma}_2 = 0.0525(1 - 0.0525)(0.5311) \approx 0.0264. \quad (13)$$

Therefore, a 10-minute decrease in driving travel time increases the probability of choosing automobile transportation by approximately 0.0264, or 2.64 percentage points.

For the probit model, the index is

$$-0.0644 + 0.3000(-5) = -1.5644. \quad (14)$$

The probit marginal effect is

$$\phi(-1.5644)(0.3000) \approx 0.1173(0.3000) = 0.0352. \quad (15)$$

Thus, the probit marginal effect is approximately 0.0352, or 3.52 percentage points. The probit estimate is slightly larger than the logit estimate.